

Predictive Forecasting of Care Load & Placement Demand

HHS Unaccompanied Alien Children (UAC) Program Dataset

Abstract

This project develops a short-term forecasting framework for children in HHS care and discharge demand using historical HHS Unaccompanied Alien Children program observations. The analysis includes data cleaning, exploratory analysis, feature engineering, chronological train/test evaluation, baseline and machine-learning forecasting, uncertainty estimation, and a historical capacity-stress indicator. Among the tested models, the Naive forecast achieved the lowest held-out error, with MAE 10.24, RMSE 14.05, and MAPE 0.454%.

1. Introduction

Care-load forecasting can support operational planning by providing an estimate of near-term population levels and discharge demand. The objective of this project is to forecast HHS care load, examine transfer-versus-discharge pressure, quantify uncertainty, and provide an early warning indicator for unusually high care loads.

2. Dataset

The dataset contains 720 valid observations after removing 450 completely blank rows. The observed period runs from 2023-01-12 through 2025-12-21. Variables include children apprehended and placed in CBP custody, children in CBP custody, transfers out of CBP custody, children in HHS care, and children discharged from HHS care.

3. Data Preparation

Dates were converted to datetime and observations were sorted chronologically. Numeric fields were converted to numeric types, including removal of comma separators from HHS care values. Duplicate dates were checked. The date range contains fewer observations than calendar days, indicating irregular reporting. Missing calendar dates were therefore not filled by artificial interpolation.

4. Exploratory Data Analysis

HHS care load shows major structural changes: it rose to a maximum of 11,516 in December 2023, declined substantially during 2024, reached approximately 2,000 during 2025, and then recovered gradually. Transfers and discharges also declined sharply during early 2025. Net pressure, defined as transfers minus discharges, had a mean of -44.74; 475 observations were negative, 238 positive, and 7 zero.

5. Feature Engineering

The forecasting dataset includes 1-, 7-, and 14-observation lags, 7- and 14-observation moving averages, lagged transfer and discharge variables, and the reporting gap in days. Observation-based lags were used because the source data is irregularly reported.

6. Forecasting Models

Four models were evaluated: a Naive persistence baseline, a 7-observation moving average, ARIMA(1,1,1), and Random Forest regression. Evaluation used a chronological 80/20 split, with older observations used for training and the latest observations reserved for testing.

7. Model Results

The Naive model clearly outperformed the other tested models on the held-out test period. Its MAE was 10.24, compared with 24.65 for the moving average, 54.19 for Random Forest, and 235.21 for ARIMA(1,1,1). The corresponding MAPE values were 0.454%, 1.101%, 2.558%, and 11.114%.

8. Forecast and Uncertainty

The final short-term forecast uses the latest observed HHS care value as the point forecast. Approximate 95% prediction intervals are calculated from the historical variability of one-step changes and widen with the square root of the selected observation horizon. The same persistence approach is used for discharge demand. These intervals should be interpreted as approximate uncertainty bands rather than formal probabilistic forecasts.

9. Capacity Stress Indicator

Because no official operational capacity limit is supplied in the dataset, a historical 90th-percentile threshold is used as an analytical stress indicator. The threshold is 9,762 children. The latest observed HHS care load is 2,484, so the current status is classified as LOW under this historical indicator. This should not be interpreted as an official HHS capacity assessment.

10. Limitations

The primary limitations are irregular reporting dates, absence of an official capacity threshold, lack of external explanatory variables, and the strong persistence of the target series. The Naive model is effective for short-term persistence but may not respond quickly to unexpected structural changes. Future work could use walk-forward validation, calendar-aware modeling, additional operational covariates, and formally calibrated probabilistic forecasts.

11. Conclusion

The analysis demonstrates that a simple persistence forecast provides the strongest short-term performance among the tested models for this dataset. The resulting dashboard combines historical care-load trends, short-term forecasts, discharge demand, model comparison, uncertainty bands, and a historical stress indicator in a practical decision-support interface.

Table 1. Held-out test-set model comparison.

Model	MAE	RMSE	MAPE (%)
Naive	10.24	14.05	0.454
Moving Average (7)	24.65	30.52	1.101
Random Forest	54.19	79.18	2.558
ARIMA (1,1,1)	235.21	285.63	11.114